

Kevin Nam

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About Me

I am currently pursuing a combined MS/PhD in the Security Optimization Research (SOR) Lab at Seoul National University. The research topics I have been following are focused on hardware/system/AI security – actually, I would rather say **privacy**. I have been active in topics related with privacy enhancing techniques (PETs), especially on how to integrate them with applications such as Machine Learning as a Service.

I deem that privacy is a crucial factor in the following era, and is eager to contribute in related fields. My research goal is to **design efficient privacy-enhancing computing systems**. To achieve this goal, I am currently pursuing research on various PETs (Homomorphic Encryption, Multi-Party Computation, Trusted Execution Environments etc). Specifically, my research topics lie in applying non-numerical approaches (HW acceleration, compiler, algorithm related techniques) to optimize the performance and minimize the downsides of PETs.

Education

Seoul National University, BS in Electrical and Computer Engineering Mar 2014 – Feb 2020

- Left for military service (ROKAF) during 2017-2019

Seoul National University, MS/Ph.D in Electrical and Computer Engineering Mar 2020 – Feb 2026*

- * expected
- Dissertation Title (tentative): Optimizations for Fast and Precise Privacy-Preserving Machine Learning as a Service over Encrypted Data

Publications

Refereed Conference Publications

- [1] **[Security'25]** "SLOTHE: Lazy Approximation of Non-Arithmetic Neural Network Functions over Encrypted Data". Kevin Nam*, Youyeon Joo*, Seungjin Ha, and Yunheung Paek. (* co-first authors) In *USENIX Security Symposium (Security'25)*, August 2025.
(KIISE S, BK21 IF: 3, CSRankings).
- [2] **[ASPLOS'25]** "Affinity-based Optimizations for TFHE on Processing-in-DRAM". Kevin Nam, Heonhui Jung, Hyunyoung Oh, and Yunheung Paek. In the *30th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'25)*, April 2025.
(KIISE S, BK21 IF: 4, CSRankings).
- [3] **[Security'25]** "LOHEN: Layer-wise Optimizations for Neural Network Inferences over Encrypted Data with High Performance or Accuracy". Kevin Nam, Youyeon Joo, Dongju Lee, Seungjin Ha, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek. In *USENIX Security Symposium (Security'25)*, August 2025.
(KIISE S, BK21 IF: 3, CSRankings).
- [4] **[ICCAD'22]** "Accelerating N-bit Operations over TFHE on Commodity CPU-FPGA". Kevin Nam, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek. In *IEEE/ACM International Conference on Computer-Aided Design (ICCAD'22)*, November 2022.
(KIISE A, BK21 IF: 3, CSRankings).
- [5] "Area-Efficient Accelerator for the Full NTRU-KEM Algorithm". Yongseok Lee, Kevin Nam, Youyeon Joo, Jeehwan Kim, Hyunyoung Oh, Yunheung Paek. In *International Conference on Computational Science and Its Applications (ICCSA'23)*. Cham: Springer Nature Switzerland, 2023.

Refereed Journal Publications

- [6] "An Efficient Hardware/Software Co-design for FALCON on Low-End Embedded Systems". Yongseok Lee, Jonghee Youn, Kevin Nam, Heon Hui Jung, Myunghyun Cho, Jimyung Na, Jong-Yeon Park, Seungsu Jeon, Bo Gyeong Kang, Hyunyoung Oh, Yunheung Paek. In *IEEE Access*, 2024.
- [7] "MeetGo: A trusted execution environment for remote applications on FPGA". Hyunyoung Oh, Kevin Nam, Seongil Jeon, and Yeongpil Cho, Yunheung Paek. In *IEEE Access*, 2021.

Refereed Poster Publications

- [8] "Affinity-based Optimizations of Homomorphic Encryption Operations on Processing-in-DRAM". Kevin Nam, Heonhui

Jung, Hyunyoung Oh, and Yunheung Paek. In the 61th ACM/IEEE Design Automation Conference Work-in-Progress (DAC'24 WiP), 2024.

- [9] "Implementing Efficient, Precise N-bit Operations of TFHE on Commodity CPU-FPGA". Kevin Nam, Youyeon Joo, Dongju Lee, Seungjin Ha, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek. In the 59th ACM/IEEE Design Automation Conference Work-in-Progress (DAC'22 WiP), 2022.

Domestic (Only First Author Papers)

- [10] "Partially Homomorphic Encryption HW accelerator". Kevin Nam, Jiwon Chang, Myunghyun Cho, Inyoung Bang, Yunheung Paek. Annual Conference of KIPS, 2020
- [11] "Realization of Homomorphic Encrypted Deep Learning Models". Kevin Nam, Myunghyun Cho, Hyunjun Kim, Yunheung Paek. Annual Conference of KIPS, 2021, (**Undang Academic Award**)
- [12] "Side-Channel Attacks on Homomorphic Encryption and Their Mitigation Methods". Kevin Nam, Youyeon Joo, Yunheung Paek. Annual Symposium of KIPS, 2023 (**NIPA Director Award**)
- [13] "Quantization of Non-Arithmetics to Optimize CNNs over TFHE". Kevin Nam, Heonhui Jung, Dongju Lee, Yunheung Paek. Annual Symposium of KIPS, 2024
- [14] "Intermediate Data Guided Approximation for Fast and Accurate Encrypted Neural Networks". Kevin Nam, Youyeon Joo, Seungjin Ha, Yunheung Paek. Annual Conference of KIPS, 2024.

Patents

- [15] "METHOD FOR PROCESSING HOMOMORPHIC CIPHERTEXT AND ELECTRONIC APPARATUS". Yunheung Paek, Heonhui Jung, Kevin Nam, Seungjin Ha, Junbum Shin, Inkwan Yu, Sunchul Jung, Jungjoo Seo. Feb 2025. US Patent No. 19/052442, Applied
- [16] "METHOD FOR PROCESSING HOMOMORPHIC CIPHERTEXT AND ELECTRONIC APPARATUS". Yunheung Paek, Heonhui Jung, Kevin Nam, Seungjin Ha, Junbum Shin, Inkwan Yu, Sunchul Jung, Jungjoo Seo. Jan 2025. KR Patent No. 10-2025-0008424, Applied
- [17] "METHOD AND APPRATUS FOR COMPUTING ENCRYPTED DATA USING MULTI-HOMOMORPHIC ENCRYPTION SYSTEM". Yunheung Paek, Kevin Nam, Youyeon Joo. Nov 2024. KR Patent No. 10-2024-0153707, Applied
- [18] "TEE ENVIRONMENT PROVIDING APPARATUS AND METHOD USING FPGA". Yunheung Paek, Hyunyoung Oh, Kevin Nam, Yeongpil Cho. Jan 2021. KR Patent No. 10-2021-0006281, Applied

Research Experience (w. fundings, PI : Yunheung Paek)

Zero-Error Privacy-Preserving Neural Networks funded by NRF, South Korea	Oct 2023 – present
• Research on desining efficient and accurate neural network services over encrypted data • Integrate non-mathematical methods (e.g., compiler) to enhance the performance of cryptographic backends • Currently participating as the FHE project leader	
Secure FHE Key Management System funded by CryptoLab, South Korea	Mar 2023 – April 2024
• Designed efficient TEE-based FHE key management system within SGX restrictions • Participated as the TEE software engineer	
Neural Network Program to FHE Transpiler funded by ETRI, South Korea	Oct 2023 – Nov 2023
• Designed a transpiler that changes tensorflow/Pytorch programs into FHE programs • Developed the FHE routines that corresponds to the kernels used in tensorflow (e.g., convolution, activations)	
PEC Platform funded by the National Intelligence Service, South Korea	Mar 2022 – Nov 2023
• Designed a Privacy-Enhancing-Computation platform using multiple types of privacy-enhancing techniques (PETs) • Developed a framework that analyzes normal codes and suggest appropriate PETs and generates a secure code • Participated as the project leader (designing the dataflow of the platform/framework)	
FPGA Accelerator for CKKS funded by CryptoLab, South Korea	Mar 2022 – Nov 2022
• Designed an FPGA-based CKKS Accelerator for its bootstrapping algorithm • Participated as the algorithm analyzer - designed the overall operation mapping and pipeline	
FPGA Accelerator for TFHE funded by the National Intelligence Service	Mar 2021 – Nov 2021
• Designed an FPGA-based TFHE accelerator exploiting the double parallelism of TFHE • Developed a methodology to find the optimal setup of parallelism considering the system's memory bandwidth	

- Participated as the sole implementation researcher, from algorithm analysis to system integration

Honors & Awards

Undang Academic Award <i>from KIPS, Best Graduate Student Paper Award</i>	Dec 2021
NIPA Director Award <i>from ASK 2023</i>	Jun 2023
Award for Excellence in Teaching Assistant , <i>from Seoul National University</i> <i>Logic circuit Design Course</i>	Jan 2024
Scholarship BK 21 + Scholarship <i>by the Ministry of Education of Korea</i>	Sep 2020 - Feb 2021 Mar 2023 - present

Activities

EuroSys'25 <i>Shadow Program Committee</i>	2025
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Teaching (Assistant) Experience

Seoul National University <i>Logic Circuit Design Course, 430.201A (TA)</i>	Sep 2023 - Dec 2023 Sep 2024 - Dec 2024
<i>Data Security & Privacy, 430.658 (TA)</i>	Sep 2024 - Dec 2024

Skills

Languages: English, French, Korean

Programming Languages: C/C++, Python, HDL, JAVA, etc

Tools/Frameworks: Vitis, Vivado, Xcellium, Synopsys DC, Tensorflow, Pytorch, Docker, Git, LLVM, etc

Privacy Enhancing Technique Tools: MS SEAL, LiberateFHE, HEonGPU, OpenFHE, HeaaN, TFHE-rs, OpenCheetah, and much more